Linear Algebra
Assignment 1

Deadline: 22 October 2026

Instructions

- Submit handwritten solutions. Calculators are not allowed.
- Complete the assignment by yourself, without using AI.
- Scan your solutions and submit them as a single PDF to the Moodle course.
- Name your submission file with your roll number: `<roll-number>.pdf`.
- The deadline is **22 October 2026**. No extension of the deadline will be granted.

Reference: Lecture notes available on Moodle. Exercise numbers and references below follow the lecture notes.

Chapter 2: Vectors and the Geometry of Space

Exercise 2.19. Let $u = (2, -1, 3)$ and $v = (0, 4, -2)$. Compute $u + v$, $u - v$, $3u$, and $\|u\|$.

Exercise 2.20. Decide whether $b = (5, 5)$ lies in $\operatorname{span}\{(1, 2), (3, 1)\}$, and if so find the coefficients.

Exercise 2.21. For which values of t does $\operatorname{span}\{(1, 3), (2, t)\}$ fail to be all of $\mathbb{R}^2$? Justify your answer geometrically as well as algebraically.

Exercise 2.22. Find the angle between $(1, 0, 1)$ and $(0, 1, 1)$. *Hint: both have the same length, which simplifies the arithmetic.*

Exercise 2.23. Definition (orthogonal projection). For $a \in \mathbb{R}^n$ with $a \neq \mathbf{0}$, the orthogonal projection of $b \in \mathbb{R}^n$ onto the line $\operatorname{span}\{a\}$ is

$$\operatorname{proj}_a(b) = \frac{\langle b, a \rangle}{\langle a, a \rangle}\, a, \qquad \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i.$$

It is the vector p on this line for which $b - p$ is perpendicular to a.

Project $b = (2, 3)$ onto the line spanned by $a = (1, -1)$, and verify that the error is perpendicular to a.

Exercise 2.24. Cauchy–Schwarz inequality. For all $x, y \in \mathbb{R}^n$ with the usual inner product,

$$|\langle x, y \rangle| \leq \|x\|\, \|y\|, \qquad \|x\| = \sqrt{\langle x, x \rangle}.$$

Show that equality holds in Cauchy–Schwarz, $|\langle x, y \rangle| = \|x\|\, \|y\|$, precisely when x and y are parallel, meaning one is a scalar multiple of the other.

Exercise 2.25. Write the equation of the plane through $(2, 0, 1)$ with normal $(3, -1, 2)$, and find the distance from the origin to that plane.

Exercise 2.26. Take the polynomials of degree at most two, with $p + q$ and λp defined coefficientwise as in Example 2.4. Compute $2(1 + t) - 3(t - t^2)$, and decide whether $1 + t^2$ lies in $\operatorname{span}\{1 + t,\ t + t^2\}$.

Exercise 2.27. Prove that the projection of Definition 2.13 deserves its name: for p the projection of b onto the line spanned by a, show that $\|b - p\| \le \|b - ta\|$ for every scalar t, with equality only when $ta = p$. *Hint: expand $\|b - ta\|^2$ as a quadratic in t.*

Chapter 3: Linear Systems and Elimination

Exercise 3.18. Solve by elimination, showing the augmented matrix at each step:

$$2x + y - z = 0, \qquad x - y + 2z = 9, \qquad 3x + 2y + z = 7.$$

Verify your answer in all three original equations.

Exercise 3.19. For each system, decide without fully solving it whether there are no solutions, one, or infinitely many, and say how you know.

(a) $x + y + z = 1, \quad 2x + 2y + 2z = 3.$

(b) $x + y = 1, \quad x - y = 3, \quad 2x = 4.$

(c) $x + y + z = 0, \quad x - z = 0.$

Exercise 3.20. Find the complete solution set of $x_1 + 2x_2 + 3x_3 + 4x_4 = 5$, written in the particular-plus-homogeneous form of Theorem 3.14. How many free parameters are there, and why does Theorem 3.10 predict that number?

Exercise 3.21. For which values of λ does the system

$$x + 2y = 1, \qquad 3x + \lambda y = 2$$

have a unique solution, and for which does it have no solution? One of the three outcomes of Theorem 3.2 never occurs here; identify it and explain why. Draw the cases you do find.

Exercise 3.22. Let A be 3×5. Explain why $Ax = 0$ must have a nonzero solution, and give the smallest possible number of free parameters. Then construct an A that attains it.

Exercise 3.23. Show that if u and v both solve $Ax = b$, then $\lambda u + (1 - \lambda)v$ solves it for every $\lambda \in \mathbb{R}$, but $u + v$ solves it only when $b = 0$. Interpret both statements using Figure 3.4.

Exercise 3.24. Bring the matrix

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 8 \\ 1 & 2 & 2 & 5 \end{pmatrix}$$

to reduced row echelon form. Identify the pivot columns and the free columns, and state the rank.

Exercise 3.25. The same $n \times n$ matrix A must be used to solve $Ax = b$ for k different right-hand sides. Strategy one runs elimination from scratch each time, at about $\frac{2}{3}n^3$ operations per solve. Strategy two computes A^{-1} once, at about $2n^3$ operations, and then multiplies, at about $2n^2$ operations per solve. For large n, at roughly what k does strategy two overtake strategy one? Library routines do neither: they factor A once at about $\frac{2}{3}n^3$ and then solve each system at about $2n^2$. Explain why that beats both.

Exercise 3.26. Theorem 3.9 asserts uniqueness for the *reduced* form only. For

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix},$$

find two different sequences of elementary row operations reaching two *different* matrices in row echelon form in the sense of Definition 3.6, and then check that reducing each one further gives the same result. Where in the definition is the freedom that allows this?

Chapter 4: Matrices, Inverses and Factorisation

Exercise 4.27. Write down the 3×3 elementary matrices for the operations $R_1 \leftrightarrow R_3$, $R_2 \to 4R_2$ and $R_3 \to R_3 - 2R_1$, and verify in each case that the inverse is the elementary matrix of the reversed operation.

Exercise 4.30. Find the LU factorisation of $A = \left(\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 5 & 4 \\ 3 & 8 & 8 \end{smallmatrix}\right)$ and use it to solve $Ax = (4,\, 11,\, 19)$. Then solve $Ax = (1,\, 1,\, 1)$ reusing the same factorisation, and count how many arithmetic operations the second solve required compared with the first.

Exercise 4.32. Give a 2×3 matrix A and a 3×2 matrix B with $AB = \mathbf{1}_2$, and compute BA. Explain why no choice of A and B of these shapes can also give $BA = \mathbf{1}_3$. *Hint: if $BA = \mathbf{1}_3$ then $Ax = 0$ forces $x = 0$; now apply Proposition 3.13 to A, which is wide.*